

FinTrust Digital Bank – Data Quality Documentation

1. Introduction

The FinTrust dataset contains two datasets: a Customer dataset and a Transaction dataset. The data was reviewed and cleaned before carrying out further analysis using Python, Power BI and SQL.

The main purpose of the data quality checks was to identify missing values, duplicates, incorrect data types, inconsistent values, unusual values and possible outliers, and to make sure that the Customer and Transaction datasets could be linked correctly.

2. Dataset Overview

Customer Dataset

- Number of records: 1,500
- Number of columns: 12
- Customer_ID: 1,500 unique IDs
- No duplicate records were identified.
- No missing values were identified.

Transaction Dataset

- Number of records: 12,000
- Number of columns: 11
- Transaction_ID: 12,000 unique IDs
- No duplicate records were identified.
- 96 missing Device_Type values were identified.
- 96 missing Location values were identified.

3. Data Types

All columns in both the Customer and Transaction datasets were checked in Power Query to confirm that they had an appropriate data type for their intended use.

The following data type corrections were identified and made:

1. **Age** – changed to Whole Number.
2. **Tenure_Months** – changed to Whole Number.
3. **Digital_Engagement_Score** – changed to Decimal Number.

4. **Amount_NGN** – changed to an appropriate numerical format.
5. **Transaction_DateTime** – changed/validated as Date/Time.

The remaining columns were reviewed and confirmed to have appropriate data types for their intended use.

Identifier fields such as **Customer_ID** and **Transaction_ID** were retained as text/identifier fields because they are used to identify records rather than for mathematical calculations.

4. Missing Values

The Customer dataset did not have any missing values identified during the assessment.

In the Transaction dataset:

- 96 Device_Type values were missing.
- 96 Location values were missing.

Instead of removing these transactions, the missing values were replaced with "**Unknown**". This allowed the transaction records to be retained for analysis while making it clear that the original information was unavailable.

5. Duplicate Records and Unique IDs

No duplicate records were identified in either dataset.

The Customer dataset contained 1,500 unique Customer_IDs across 1,500 records.

The Transaction dataset contained 12,000 unique Transaction_IDs across 12,000 records.

This means the IDs could be used to identify individual customer and transaction records.

6. Value Consistency

The values in the Customer and Transaction datasets were reviewed for inconsistent categories or formats.

No major inconsistent values were identified.

The Monthly_Income_Band column required additional attention. The values were valid and consistently formatted, but the categories were not arranged in a logical income order. A

custom sort order was therefore used during the analysis so that the income bands could be displayed correctly.

7. Customer and Transaction Relationship

Customer_ID was used as the common field between the Customer and Transaction datasets.

The Customer_ID values in the Transaction dataset were checked against the Customer dataset, and all transaction Customer_IDs matched an existing customer.

This means the two datasets could be joined for analysis without unmatched customer records being identified.

8. Outlier Checks

Potential outliers were reviewed using the $1.5 \times \text{IQR}$ method.

Digital Engagement Score

Four potential statistical outliers were identified:

- 18.0
- 17.3
- 17.1
- 16.7

Although these values were identified statistically as potential outliers, they were still within the valid range of 10–100.

The values were therefore retained because there was no evidence that they were incorrect.

Tenure

No statistical outliers were identified for Tenure_Months using the $1.5 \times \text{IQR}$ method.

The calculated lower bound was -46.00 and the upper bound was 146.00.

Age

No statistical outliers were identified for Age using the $1.5 \times \text{IQR}$ method.

Transaction Amount

There were 1,474 potential statistical outliers in Amount_NGN based on the $1.5 \times \text{IQR}$ method.

These values were reviewed and appeared to represent plausible transaction amounts. They were therefore retained because there was no evidence that they were data-entry errors.

9. Main Cleaning Activities

The main cleaning activities carried out were:

1. Checked the number of records and columns in both datasets.
2. Checked for missing values.
3. Replaced missing Device_Type values with "Unknown".
4. Replaced missing Location values with "Unknown".
5. Checked for duplicate records.
6. Checked that Customer_ID and Transaction_ID values were unique.
7. Checked and corrected data types in Power Query.
8. Reviewed category consistency.
9. Applied a logical sort order to Monthly_Income_Band during analysis.
10. Checked the relationship between the Customer and Transaction datasets.
11. Checked potential outliers using the $1.5 \times \text{IQR}$ method.
12. Reviewed potential outliers and retained values that appeared to be valid.

10. Data Quality Issues and Their Impact

Most of the issues identified did not require records to be removed.

The missing Device_Type and Location values could affect analysis involving devices and locations, so they were replaced with "Unknown" rather than deleting the affected transactions.

Incorrect data types could affect calculations and analysis, so the relevant columns were changed to appropriate formats.

The Monthly_Income_Band ordering could affect the way results were displayed, so a custom sort order was applied.

The potential outliers were reviewed rather than automatically removed because statistical unusualness does not necessarily mean that a value is incorrect.

11. Data Limitations

The transaction data covers the period from 1 January 2026 to 3 December 2026. December is therefore not a complete month.

This should be considered when interpreting trends or transaction activity over time.

Also, the identification of a statistical outlier does not automatically mean that the value is invalid. The values were therefore reviewed before deciding whether any action was required.

12. Overall Assessment

The data was reviewed and cleaned before the business analysis was carried out.

The main issues identified were missing Device_Type and Location values, incorrect data types in some fields, the ordering of Monthly_Income_Band categories, and potential statistical outliers.

The missing values were handled, data types were corrected, the customer and transaction relationship was validated, and potential outliers were reviewed.

After these checks and cleaning activities, the datasets were suitable for the analysis carried out in Python, Power BI and SQL.